EEMT 5510
Engineering Economics and Cost Management

Chapter 6
Project comparison and selection

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Comparison and selection among alternatives

- Example 1:
  - A technology company can have multiple investment opportunities: expand its current technology business, build autonomous cars (driverless), buy the shares of other companies, etc.
- Example 2:
  - You need a smart phone. Different brands to consider: iPhone, Samsung, Huawei, etc.

Comparison and selection among alternatives

- There are multiple alternatives for a project
  - Mutually exclusive alternatives: only one should be selected
    - Doing nothing is also an alternative
    - Example: Our budget can only afford one smart phone.
- Considerations
  - Investments
  - Revenue and profits
  - Useful lives
  - Criteria

The study (analysis) period and useful lives

- Useful life of an alternative: the time it will last.
  - Case 1. Equal useful lives for all alternatives
  - Case 2. Unequal useful lives
- Rule: Compare alternatives over the same time period, which is called study period.
- The choice of study period depends on the problem (in practice there is often a natural choice).

Investment alternatives

- Investment alternatives:
  - With initial capital investment that produce positive cash flows from revenue, savings through reduced costs, or both.
- Implicit alternative: MARR investment.
- If none of the alternatives are better than the MARR investment, then none of them should be chosen, and we should invest elsewhere at MARR instead.

Cost alternatives

- Cost alternatives:
  - all negative cash flows except for a possible cash-flow from disposal of assets at the end of the project's useful life.

- Example:
  - In order to test a new drug, a pharmaceutical company has to buy new equipment: initial purchase cost, annual maintenance cost, salvage value when the test ends.

Basic concepts in comparing alternatives

- Focus on the difference
  - The alternative that requires the minimum investment of capital and produces satisfactory functional results will be chosen, unless the incremental capital associated with an alternative having a larger investment can be justified with respect to its incremental benefits.

Base alternative

- The base alternative is the one that is acceptable and needs the minimum amount of investment
  - Acceptable: evaluated by MARR
    - Equivalence of profit >0, or IRR(ERR)>MARR
  - A base alternative is often identified first
- For other alternatives
  - Can the extra investment be justified?

The study (analysis) period

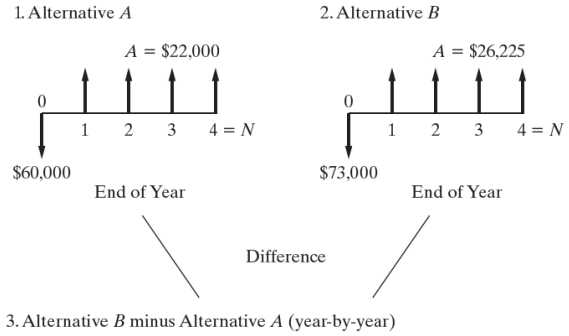
- Useful lives are the same for all alternatives and equal to the study period.
- Useful lives are unequal among the alternatives, and at least one does not match the study period.

Example 6.1: Investment alternatives

- Consider the following two projects, both of which will run for 4 years:

	Alternative A	Alternative B	Δ(B-A)
Capital investment	60,000	73,000	13,000
Annual revenue	22,000	26,225	4,225

- MARR = 10%



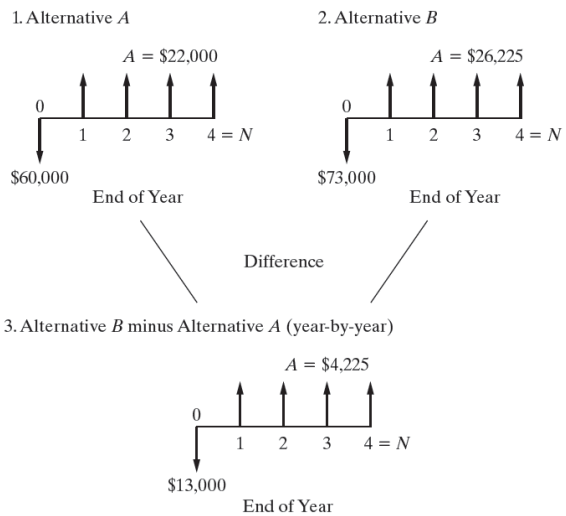
Example 6.1: Investment alternatives

- If MARR = 10% and we use the PW decision rule

$$PW_A(10\%) = -60{,}000 + 22{,}000(P/A, 10\%, 4) = 9{,}738$$
$$PW_B(10\%) = -73{,}000 + 26{,}225(P/A, 10\%, 4) = 10{,}131$$
$$PW_{\Delta(B-A)}(10\%) = -13{,}000 + 4{,}225(P/A, 10\%, 4) = 393$$

- The first two equations imply that both A and B are economically justified projects.
- The last equation shows that Project B is preferred over Project A.

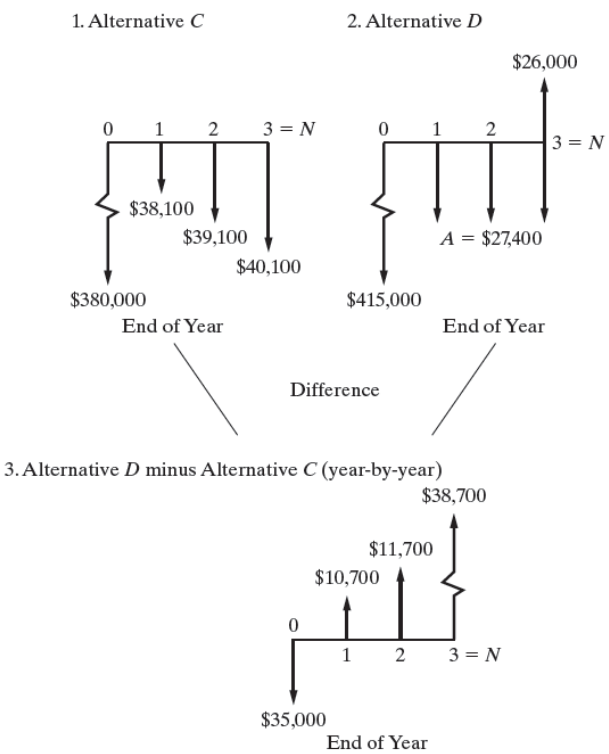
$$PW_{\Delta(B-A)}(10\%) = PW_B(10\%) - PW_A(10\%)$$



Example 6.2: Cost alternatives

- Alternatives C and D are two mutually exclusive cost alternatives with estimated net cash flows, as shown, over a three-year life.

End of Year	Alternative C	Alternative D	Δ(D-C)
0	-380,000	-415,000	-35,000
1	-38,100	-27,400	10,700
2	-39,100	-27,400	11,700
3	-40,100	-27,400	12,700
3[a]	0	26,000	26,000



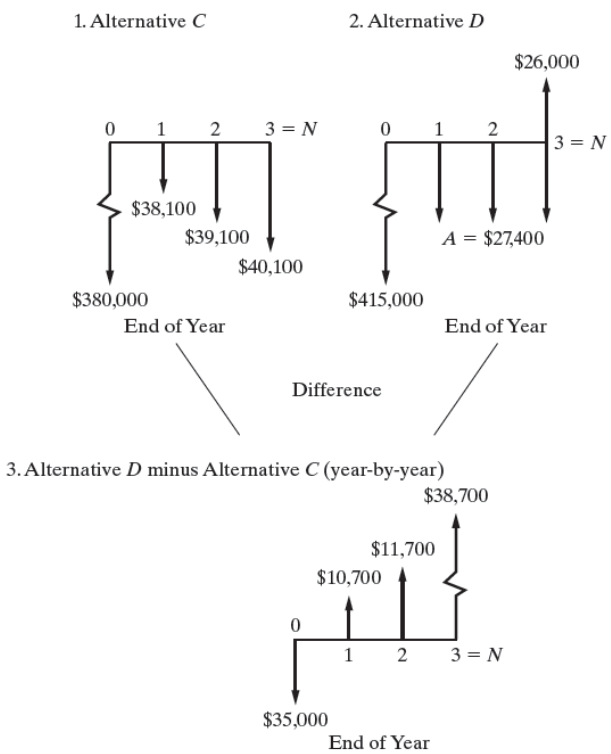
- MARR = 10%

Example 6.2: Cost alternatives

- Alternatives C and D are two mutually exclusive cost alternatives with estimated net cash flows, as shown, over a three-year life.

$$PW_C(10\%) = -477{,}077$$
$$PW_D(10\%) = -463{,}607$$
$$PW_{\Delta(D-C)}(10\%) = 13{,}470$$

- Alternative D is preferred to C because it has the less negative PW (minimizes costs).



Equivalent-worth methods

- Determine the equivalent worth of each alternative based on total investment at *i* = MARR.
- For investment alternatives, the one with the greatest positive equivalent worth is selected.
- For cost alternatives, the one with the least negative equivalent worth is selected.

Example 6.3 (PW)

- Consider the following three projects, which last for 10 years. The MARR is 10%.

	Project A	Project B	Project C
Capital investment	390,000	920,000	660,000
Annual revenue	69,000	167,000	133,500

- Evaluate by PW method

$$PW_A(10\%) = -390{,}000 + 69{,}000(P/A, 10\%, 10) = 33{,}975$$
$$PW_B(10\%) = -920{,}000 + 167{,}000(P/A, 10\%, 10) = 106{,}143$$
$$PW_C(10\%) = -660{,}000 + 133{,}500(P/A, 10\%, 10) = 160{,}300$$

- $C \succ B \succ A$

Example 6.3 (AW)

- Consider the following three projects, which last for 10 years. The MARR is 10%.

	Project A	Project B	Project C
Capital investment	390,000	920,000	660,000
Annual revenue	69,000	167,000	133,500

- Evaluate by AW method

$$AW_A(10\%) = -390{,}000(A/P, 10\%, 10) + 69{,}000 = 5{,}547$$
$$AW_B(10\%) = -920{,}000(A/P, 10\%, 10) + 167{,}000 = 17{,}316$$
$$AW_C(10\%) = -660{,}000(A/P, 10\%, 10) + 133{,}500 = 26{,}118$$

- $C \succ B \succ A$

Example 6.4 (PW,AW,FW)

- A company is planning to install a new automated plastic-molding press. Four different presses are available with all useful lives as five years. MARR=10%

	Project 1	Project 2	Project 3	Project 4
Capital investment	24,000	30,400	49,600	52,000
Annual expense				
Power	2,720	2,720	4,800	5,040
Labor	26,400	24,000	16,800	14,800
Maintenance	1,600	1,800	2,600	2,000
Property taxes	480	608	992	1,040
Total annual expense	31,200	29,128	25,192	22,880

Example 6.4

- A company is planning to install a new automated plastic-molding press. Four different presses are available with all useful lives as five years. MARR=10%

	Press (Equivalent-Worth Values)			
Method	*P*1	*P*2	*P*3	*P*4
Present worth	−$142,273	−$140,818	−$145,098	−$138,734
Annual worth	−37,531	−37,148	−38,276	−36,598
Future worth	−229,131	−226,788	−233,689	−223,431

- $P4 \succ P2 \succ P1 \succ \mathrm{P3}$

Identical useful lives

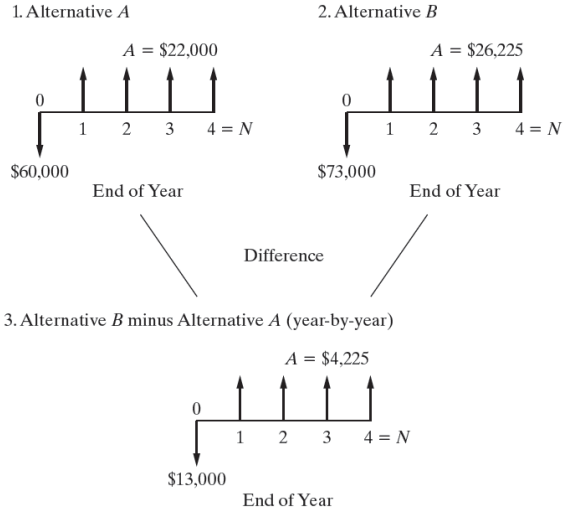
- Using equivalence value method
  - Either PW, FW, AW can be used to compare different alternatives
  - The alternative with the best equivalence value should be chosen
  - The three measurements always generate the same result
  - The identification of base alternative may not be necessary

Example 6.1 revisited

- Consider the following two projects, both of which will run for 4 years:

	Alternative A	Alternative B	Δ(B-A)
Capital investment	60,000	73,000	13,000
Annual revenue	22,000	26,225	4,225

- MARR = 10%



Example 6.1 revisited (PW/IRR)

- By the PW decision rule, we have
  - $PW_A(10\%) = 9{,}738$
  - $PW_B(10\%) = 10{,}131$
- By the IRR decision rule, we have
  - $IRR_A = 17.3\%$
  - $IRR_B = 16.3\%$
- Thus, both projects are economically justified. However, it is not clear which is better:
  - $PW_A(10\%) \leq PW_B(10\%)$ but $IRR_A \geq IRR_B$.

Recall: The Internal Rate of Return (IRR) method

- Given a cash flow, the internal rate of return (IRR) is the interest rate that makes the PW of all cash outflow equal to the PW of all cash inflow
- A project is acceptable only if IRR≥MARR
- Mathematically, *i'* is the IRR if it satisfies

$$\sum_{k=0}^{N} R_k \times (P/F, i'\%, k) = \sum_{k=0}^{N} E_k \times (P/F, i'\%, k)$$

  - R_k: total revenue of the k-th period;
  - E_k: total expenditure of the k-th period;
  - *N*: number of periods in the planning horizon

Correct usage of IRR

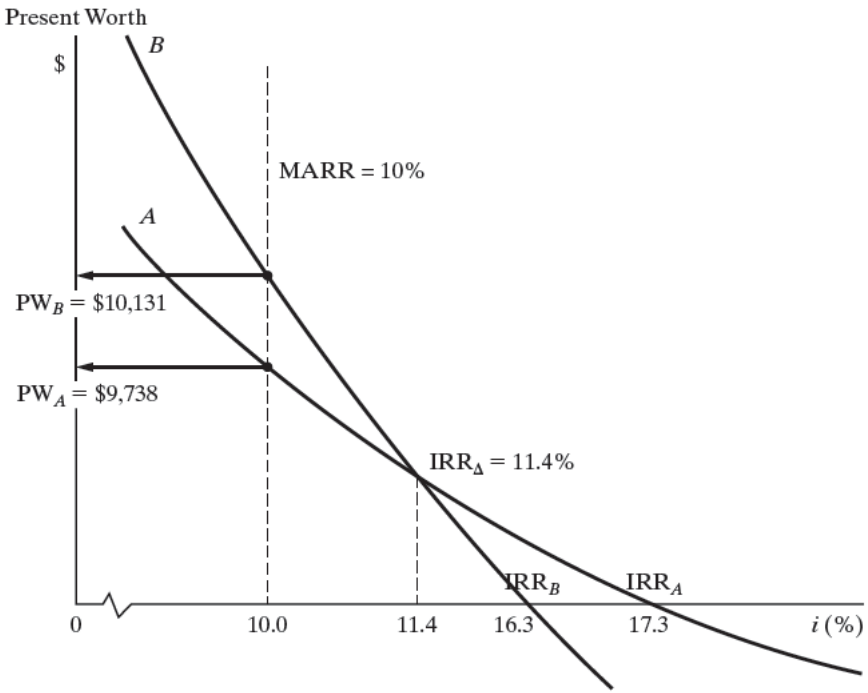
- IRR, in its simple usage, can only tell whether an alternative is acceptable or not
- The IRR of the **incremental net cash flow** can be used to compare alternatives
  - The additional investment is justified only if the IRR for the incremental net cash flow is greater than MARR

Example 6.1 revisited

- Project B requires an additional $13,000 in investment and produces an additional return of $4,225 in each year over Project A. (Why?)
- The present worth of the incremental cash flows at MARR = 10% is $393 \geq 0$
- The IRR of the incremental cash flows
  - $IRR_{\Delta} = 11.4\% \geq MARR$
- Conclusion: The use of the additional $13,000 for Project B is justified, and hence Project B is preferred over Project A.

Example 6.1 revisited

- Illustration of the ranking error in studies using the IRR Method

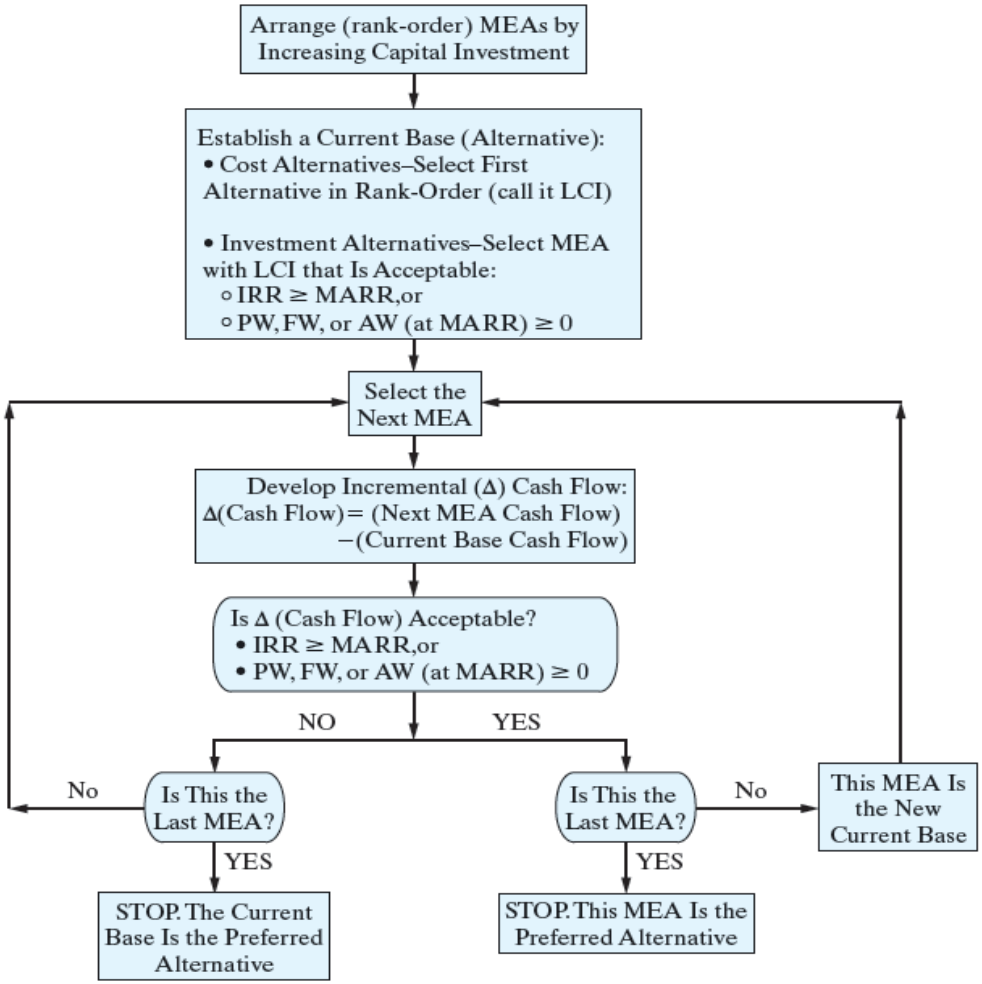


- We cannot base our choice on maximizing IRR because the actual rate is different from IRR, why?

Rate-of-return methods

- Three guidelines are applicable to rate-of-return methods:
  - Each increment of capital must justify itself by producing a sufficient rate of return (greater than or equal to MARR) on that increment.
  - Compare a higher investment alternative against a lower investment alternative only when the latter is acceptable. The difference between the two alternatives is usually an *investment alternative* and permits the better one to be determined.
  - Select the alternative that requires the largest investment of capital, as long as the incremental investment is justified by benefits that earn at least the MARR. This maximizes equivalent worth on total investment at i = MARR.

Incremental analysis procedure



MEA: mutually exclusive alternative
LCI: least capital investment

Incremental analysis procedure

- Step 1. Order the alternatives by increasing capital investment.
- Step 2. Find a base alternative (current best alternative)
  - Investment alternatives: the first acceptable alternative in the ordered list.
  - Cost alternatives: the first alternative in the ordered list (the one with the least capital investment).
- Step 3. Evaluate the difference between the next alternative and the current best alternative.
  - If the incremental cash flow is acceptable, choose the next alternative as the current best alternative.
  - Otherwise, keep the current best alternative and drop the next alternative from further consideration.
- Step 4. Repeat Step 3 until the last alternative is considered. Select the current best alternative as the best one.

Example 6.5

- Six mutually exclusive alternatives for a small investment project, using the IRR method. The useful life of each alternative is 10 years, and the MARR is 10% per year.

	A	B	C	D	E	F
Investment	900	1500	2500	4000	5000	7000
Annual Income	150	276	400	925	1125	1425

Example 6.5

- Calculate IRR for each alternative
  - Alternative C is not acceptable (why?)
  - Alternative D has the largest IRR.
    - Should we choose this?
- Choose A as the base alternative
  - Acceptable and with the least investment

	A	B	C	D	E	F
Investment	900	1500	2500	4000	5000	7000
Annual Income	150	276	400	925	1125	1425
IRR(%)	10.6	13.0	9.6	19.1	18.3	15.6

Example 6.5

- Should we accept Alternative B?
  - Incremental cash flow **B−A**: {−600, 126,…,126}
  - IRR for the incremental cash flow: 16.4%
  - Alternative B is better than A, becomes the current best
- Should we accept Alternative D?
  - Incremental cash flow **D−B**: {−250, 649,…,649}
  - IRR for the incremental cash flow: 22.6%
  - Alternative D is better than B, becomes the current best

	A	B	C	D	E	F
Investment	900	1500	2500	4000	5000	7000
Annual Income	150	276	400	925	1125	1425

Example 6.5

- Should we accept Alternative E?
  - Incremental cash flow **E−D**: {−1000, 200,…,200}
  - IRR for the incremental cash flow: 15.1%
  - Alternative E is better than D, becomes the current best
- Should we accept Alternative F?
  - Incremental cash flow **F−E**: {−2000, 300,…,300}
  - IRR for the incremental cash flow: 8.1% < MARR
  - Alternative F cannot be justified

	A	B	C	D	E	F
Investment	900	1500	2500	4000	5000	7000
Annual Income	150	276	400	925	1125	1425

Example 6.5

- Alternative E should be selected by incremental IRR analysis

Increment Considered	A	$\Delta(B-A)$	$\Delta(D-B)$	$\Delta(E-D)$	$\Delta(F-E)$
Δ Capital investment	\$900	\$600	\$2,500	\$1,000	\$2,000
Δ Annual revenues less expenses	\$150	\$126	\$649	\$200	\$300
IRR_{Δ}	10.6%	16.4%	22.6%	15.1%	8.1%
Is increment justified?	Yes	Yes	Yes	Yes	No

- How about PW method under MARR?
  - Alternative E has the largest PW, will be selected

Incremental analysis procedure

- Instead of PW, one can also use FW, AW or IRR. In all cases, the incremental cash flows must be properly identified.
- In essence, we have the following rules for selecting between two projects A and B:
  - PW Selection Rule: If $PW_{(A-B)}(MARR) \geq 0$, then A is preferred over B; otherwise, B is preferred over A.
  - IRR Selection Rule: If $IRR_{(A-B)} \geq MARR$, then A is preferred over B; otherwise, B is preferred over A.

Example 6.6

- An equipment is needed and 4 candidates are considered
  - Useful life for 5 years, MARR=20%
- Try to minimize cost
  - D1 is selected as the base alternative
  - No alternatives can be pre-eliminated at this stage

	D1	D2	D3	D4
Investment	100000	140600	148200	122000
Annual expense	29000	16900	14800	22100
Salvage value	10000	14000	25600	14000

Example 6.6

Increment Considered	$\Delta(D4 - D1)$	$\Delta(D2 - D4)$	$\Delta(D3 - D4)$
Δ Capital investment	\$22,000	\$18,600	\$26,200
Δ Annual expense (savings)	6,900	5,200	7,300
Δ Market value	4,000	0	11,600
Useful life (years)	5	5	5
IRR_{Δ}	20.5%	12.3%	20.4%
Is increment justified?	Yes	No	Yes
$PW_{\Delta}(20\%)$	\$243	−\$3,049	\$293
Is increment justified?	Yes	No	Yes

Step 1: D4 is better than D1, so D4 becomes current best

Step 2: D2 is worse than D4, so D4 is still current best

Step 3: D3 is better than D4, so D3 becomes current best; END

The study (analysis) period

- Useful lives are the same for all alternatives and equal to the study period.
- Useful lives are unequal among the alternatives, and at least one does not match the study period.

Alternatives with unequal useful lives

- Problems with different useful lives
  - How to justify a higher PW with a longer time
  - How to justify an expensive machine that can be used for a longer time
  - Basic idea: they have to be compared under a common basis
    - Comparing PW directly for two alternatives with different useful lives is unfair
- Two types of assumptions
  - Repeatability assumption
  - Co-terminated assumption

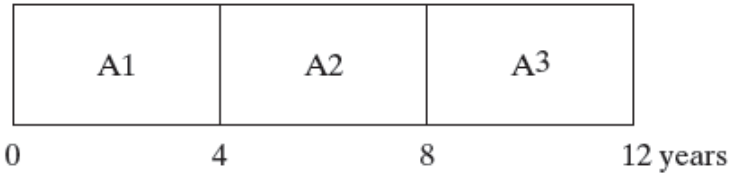
Repeatability assumption

- Assumption
  - For an alternative with a given life cycle, the alternative can be repeatedly implemented at the end of each life cycle with the same cash flow
  - For all alternatives with different useful lives, the study period is a (least) common multiple for the lives of all alternatives

Example 6.7: Repeatability assumption

	A	B
Capital investment	3500	5000
Annual cash flow	1255	1480
Useful life (years)	4	6

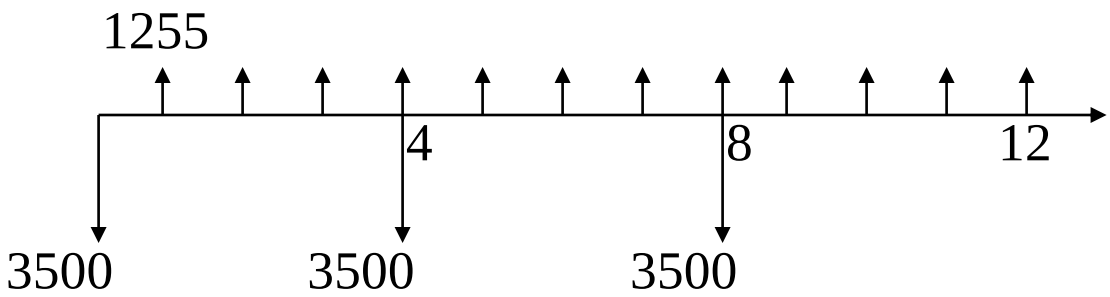
Three cycles of Alternative *A*:

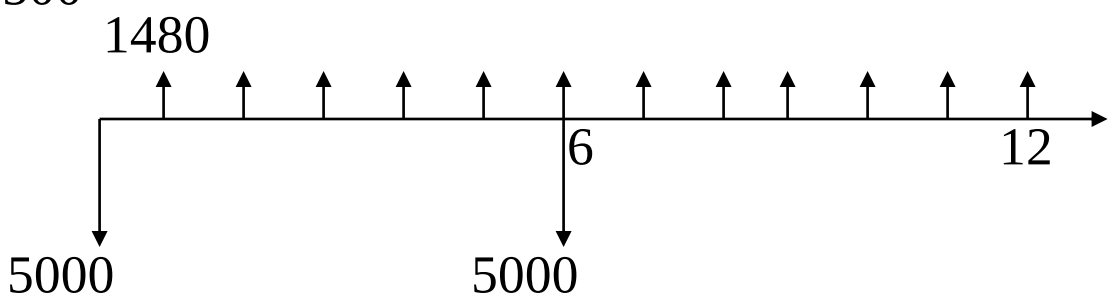


Two cycles of Alternative *B*:

Example 6.7: Repeatability assumption

	A	B
Capital investment	3500	5000
Annual cash flow	1255	1480
Useful life (years)	4	6





Example 6.7: Repeatability assumption

- PW method (PW for 12 years, MARR=10%)

  $PW_A = -3500 - 3500(P/F,10\%,4) - 3500(P/F,10\%,8)$
  $+1255(P/A,10\%,12) = 1028$

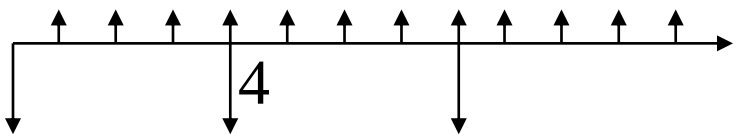
  $PW_B = -5000 - 5000(P/F,10\%,6) + 1480(P/A,10\%,12) = 2262 > PW_A$

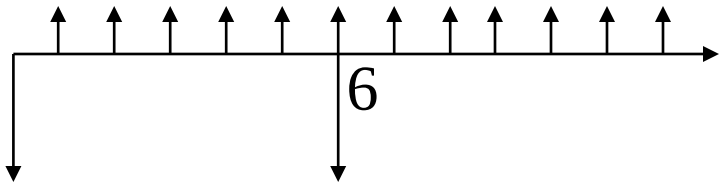
  Alternative B is better

- AW method
  - To calculate AW, based on one life cycle for each alternative is enough

  $AW_A = -3500(A/P,10\%,4) + 1255 = 151$

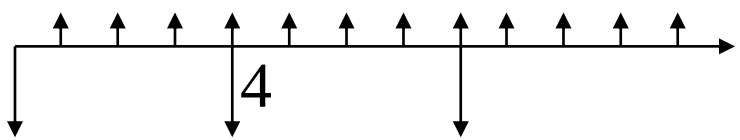
  $AW_B = -5000(A/P,10\%,6) + 1480 = 332 > AW_A$

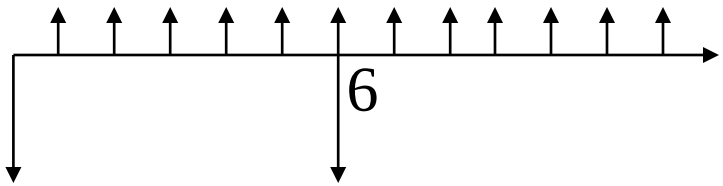




Example 6.7: Repeatability assumption

- Observation:
  - The AW of each alternative over its useful life is equal to its AW over the entire study period.
- Conclusion:
  - When repeatability assumption holds, it is sufficient to compare the AW values of the alternatives over their own useful life and select the alternative that maximizes AW.



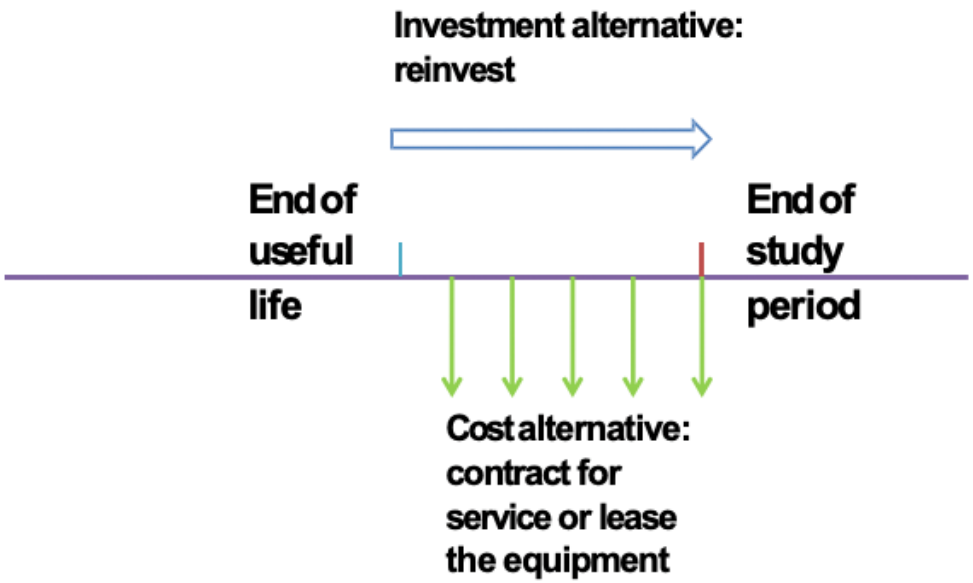


Co-terminated assumption: study period > useful life

- Assumptions that are usually made:
  - **Investment alternatives**: the cash flows are assumed to be reinvested until the end of the study period.
    - Two copper mines. One lasts for 15 years and the other for 20 years. The study period is chosen to be 20 years.
  - **Cost alternatives**: Contracting for the service or lease the needed equipment until the end of the study period. Or repeat the original alternative and use an estimated market value to truncate at the end of the study period.
    - A machine is needed for production. Type A lasts for 4 years while type B lasts for 6 years. The production period is 6 years so study period is chosen as 6 years.

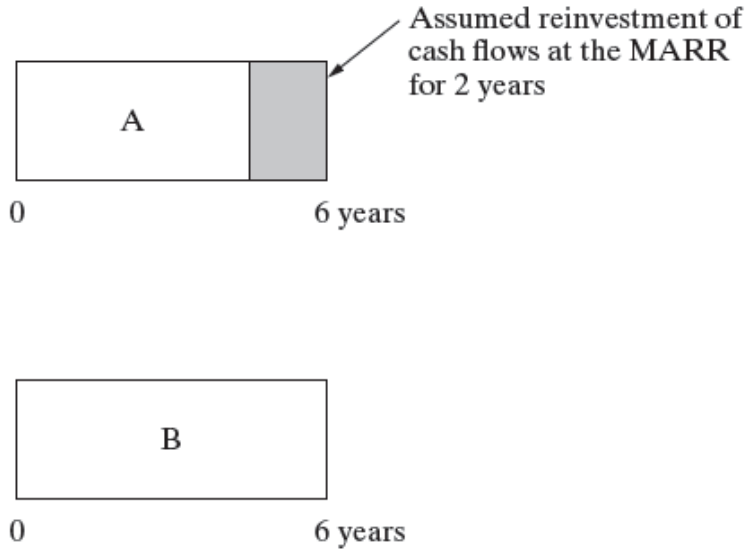
Co-terminated assumption: study period > useful life

- Assumptions that are usually made:



Example 6.8: Co-terminated assumption

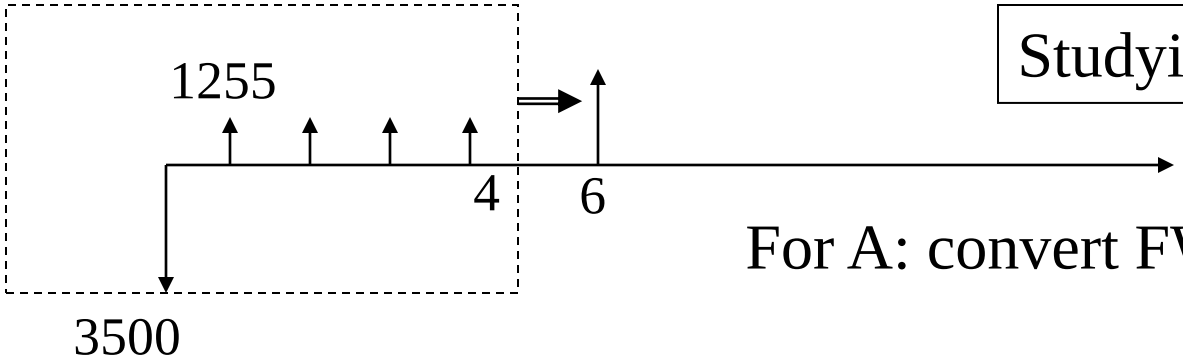
	A	B
Capital investment	3500	5000
Annual cash flow	1255	1480
Useful life (years)	4	6



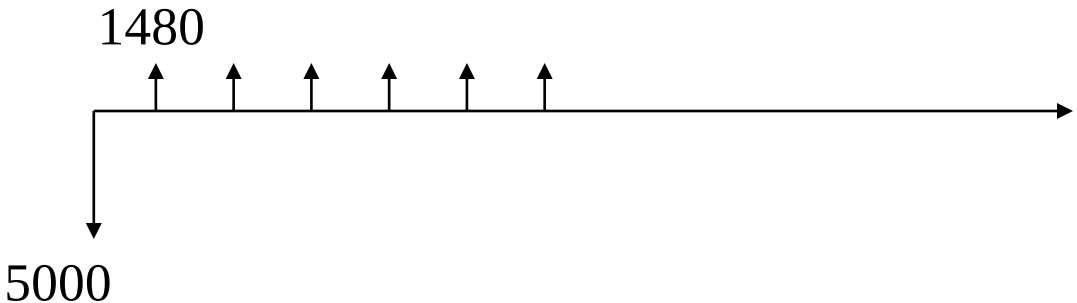
Example 6.8: Co-terminated assumption

	A	B
Capital investment	3500	5000
Annual cash flow	1255	1480
Useful life (years)	4	6

Studying period: 6 years



For A: convert FW_4 to FW_6 at MARR



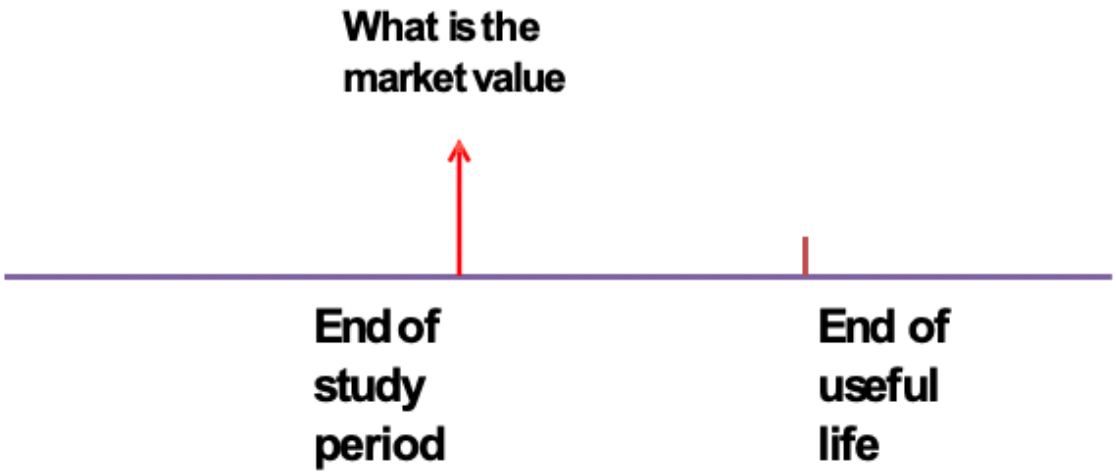
Example 6.8: Co-terminated assumption

- For an alternative with useful life (L) shorter than the studying period (S)
  - Re-invest the FW_L to the studying period FW_S at MARR
- For alternative A
  - $FW_{A,4} = -3500(F/P,10\%,4)+1255(F/A,10\%,4)$
  - $FW_{A,6}=FW_{A,4}(F/P,10\%,2)=847$
- For alternative B
  - $FW_{B,6}=-5000(F/P,10\%,6)+1480(F/A,10\%,6)=2561$
- Alternative B is preferred

Co-terminated assumption: study period < useful life

- Example: A machine is needed for production. Type A lasts for 4 years while type B lasts for 6 years. The production period is 4 years so study period is chosen as 4 years. Now study period < useful life for B.
- Truncate the alternative at the end of the study period. We need the following additional information:
  - market value of the assets at the end of the study period. We can use
    - the market value information if it is available.
    - otherwise the imputed market value.

Co-terminated assumption: study period < useful life

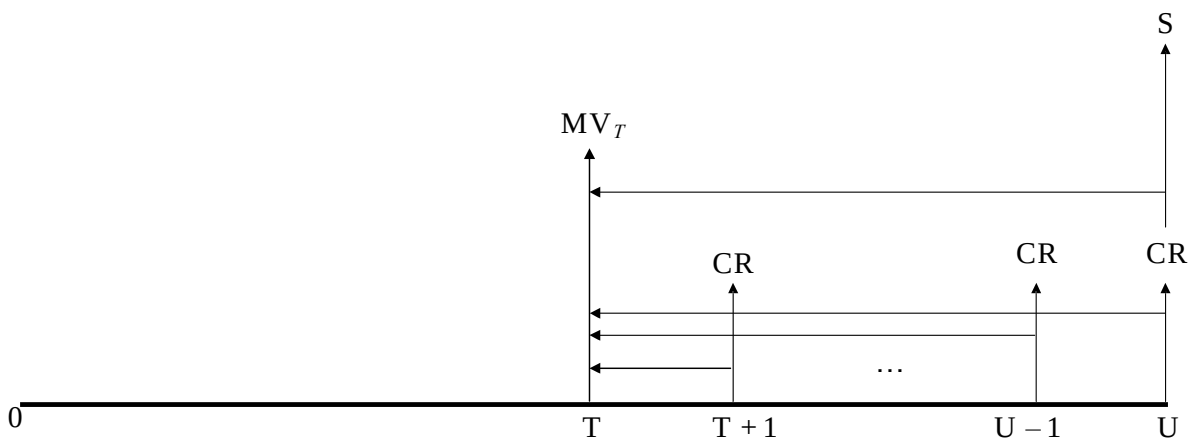


Imputed market value

- Let T denote the study period (T< useful life, U), and let MV_T denote the market value at time T.
- MV_T=PW at the end of year T of remaining capital recovery amount + PW at the end of year T of original market (salvage) value at the end of useful life.
- $MV_T = [I(A/P, MARR, U) - S(A/F, MARR, U)](P/A, MARR, U - T) + S(P/F, MARR, U - T)$
  - I be the initial investment
  - S be the salvage value of a project with a useful life of U
  - $U>T$.

Imputed market value

- $MV_T = [I(A/P, MARR, U) - S(A/F, MARR, U)](P/A, MARR, U - T) + S(P/F, MARR, U - T)$
  - *I* be the initial investment
  - *S* be the salvage value of a project with a useful life of U
  - *U>T*.



CR = I(A/P, i%, U) – S(A/F, i%, U)

CR as the lease fee

- Let's consider the following situation: a company buys a machine (initial cost I) not for production, but will lease it out. The machine is supposed to last U years with salvage value S. What should be the annual lease fee? The company looks for a rate of return equal to MARR.
- Suppose the annual lease fee is X. Then X should satisfy
  - $I = X(P/A, MARR, U) + S(P/F, MARR, U)$
    $\Rightarrow X = I(A/P, MARR, U) - S(A/F, MARR, U) = CR.$
- Hence CR can be interpreted as the annual lease fee.

Interpretation of the imputed market value

- How to determine the market value of the machine at the end of year T?
- If you have the machine, you can lease it out for $U - T$ years, charging CR every year, and recover S in the end. Now if you want to sell the machine at the end of year T, you should charge the buyer

$$CR(P/A, MARR, U - T) + S(P/F, MARR, U - T)$$

as this is the PW of future cash flows you will receive if you have the machine. Therefore

$$MV_T = CR(P/A, MARR, U - T) + S(P/F, MARR, U - T)$$

Example 6.9

- Consider a project with a capital investment of \$47,600, useful life of 9 years, annual expenses of \$1,720, and market (salvage) value of \$5,000 at the end of its useful life. What is the estimated market value of this project at the end of year 5? The MARR is 20%.

- Solution

$$MV_5 = [47{,}600(A/P,20\%,9)-5{,}000(A/F,20\%,9)](P/A,20\%,4) + 5{,}000(P/F,20\%,4)$$
$$= \$32{,}361$$

Example 6.10

- Which of the following alternatives should be selected if the study period is 5 years and the MARR is 20%?

	A	B
Capital investment	33,200	47,600
Annual expenses	2,165	1,720
Useful life (years)	5	9
Salvage value	0	5,000

Example 6.10

- Solution
  - Alternative B is identical to the project in previous example. The imputed market value of Alternative B at the end of year 5 was found to be $32,361.
  - We can use the AW method to compare the alternatives:

$$AW(20\%)_A = -33{,}200(A/P, 20\%, 5) - 2{,}165 = -13{,}267.$$

$$AW(20\%)_B = -47{,}600(A/P, 20\%, 5) - 1{,}720 + 32{,}361(A/F, 20\%, 5)$$
$$= -13{,}288.$$

  - Select A.

Equal useful lives (study period = useful life)

Investment alternatives

PW (FW, AW): Choose the one with largest positive PW (FW, AW)

or

IRR: use the incremental procedure. In step 2, choose the first acceptable alternative in the list as the current best

Cost alternatives

PW (FW, AW): Choose the one with least negative PW (FW, AW)

or

IRR: use the incremental procedure. In step 2, choose the first alternative in the list as the current best

Unequal useful lives

Repeatability

1. Study period = LCM of useful lives.
2. PW and IRR analysis can be applied to the whole study period. The most convenient approach is to use AW. For investment alternatives, choose the one with the largest positive AW. For cost alternatives, choose the one with the least negative AW.

Cotermination

1. Compare over the whole study period (PW, FW, AW, IRR).
2. For a given alternative, its useful life may be shorter or longer than the study period.
3. Useful life < study period: for investment alternatives, reinvest the cash flows; for cost alternatives, lease the equipment until the end of study period.
4. Useful life > study period: estimate the market value at the end of study period. If no market value information available, use the imputed market value technique .

Capital rationing

- Relationship among multiple alternatives/projects
  - Mutually exclusive
  - Independent
  - Contingent: project B can be selected only if project A is selected
- Total budget may be limited to fund all projects
  - A subset of projects to be selected
- Objective: Maximizing the total PW

Example 6.11

- 4 independent projects (numbers are in millions of dollars).
- Budget constraint: capital investment cannot exceed 10 million at time 0.

Project	CF_0	CF_1	CF_2	PW(10%)
A	-10	30	5	21
B	-5	5	20	16
C	-5	5	15	12
D	0	-40	60	13

Example 6.11

- Rank the projects by PW from highest to lowest. Select the project with the highest PW. If the budget left is enough, select the project with the second highest PW. Continue this process until the budget is depleted or the remaining budget is not enough for any projects.
- Applying this method, A is chosen. Is this the best solution?

Example 6.11

- x_A: either 0 or 1
  - $x_A = 0$ means we do not invest in A
  - $x_A = 1$ means we invest in A
- x_B, x_C, x_D are defined similarly
- The present worth of all selected projects is

$$21x_A + 16x_B + 12x_C + 13x_D$$

- The total capital investment is

$$10x_A + 5x_B + 5x_C$$

Example 6.11

- x_A: either 0 or 1
  - $x_A = 0$ means we do not invest in A
  - $x_A = 1$ means we invest in A
- Optimization problem

$$\begin{aligned} \max \quad & 21x_A + 16x_B + 12x_C + 13x_D \\ \text{s.t.} \quad & 10x_A + 5x_B + 5x_C \leq 10 \\ & x_A, x_B, x_C, x_D \in \{0, 1\} \end{aligned}$$

- Optimal solution:

$$x_A^* = 0, x_B^* = x_C^* = x_D^* = 1, PW^* = 41$$

Example 6.11

- Additional constraint
  - The capital investment at the end of year 1 cannot exceed 10 million
    - Optimization problem

$$\begin{aligned} \max \quad & 21x_A + 16x_B + 12x_C + 13x_D \\ \text{s.t.} \quad & 10x_A + 5x_B + 5x_C \le 10 \\ & -30x_A - 5x_B - 5x_C + 40x_D \le 10 \\ & x_A, x_B, x_C, x_D \in \{0, 1\} \end{aligned}$$

    - Optimal solution:

$$x_A^* = x_D^* = 1, \quad x_B^* = x_C^* = 0, PW^* = 34$$

  - A and D are mutually exclusive
    - $x_A + x_D \le 1$

Example 6.11

- Additional constraint
  - The capital investment at the end of year 1 cannot exceed 10 million
  - A and D are mutually exclusive
    - Optimization problem

$$
\begin{aligned}
\max \quad & 21x_A + 16x_B + 12x_C + 13x_D \\
\text{s.t.} \quad & 10x_A + 5x_B + 5x_C \le 10 \\
& -30x_A - 5x_B - 5x_C + 40x_D \le 10 \\
& x_A + x_D \le 1 \\
& x_A, x_B, x_C, x_D \in \{0, 1\}
\end{aligned}
$$

    - Optimal solution:

$$x_A^* = x_D^* = 0, \quad x_B^* = x_C^* = 1, PW^* = 28$$

Example 6.11

- Additional constraint
  - If projects A, B and C are mutually exclusive

$$x_A + x_B + x_C \leq 1$$

  - If project B can be chosen only if A has been selected

$$x_B \leq x_A$$

  - If project C is dependent on the acceptance of A or B

$$x_C \leq x_A + x_B$$

Example 6.11

- Mutually exclusive is a special case
  - Suppose A, B, C and D are mutually exclusive

$$x_A + x_B + x_C + x_D \leq 1$$

  - Optimal solution

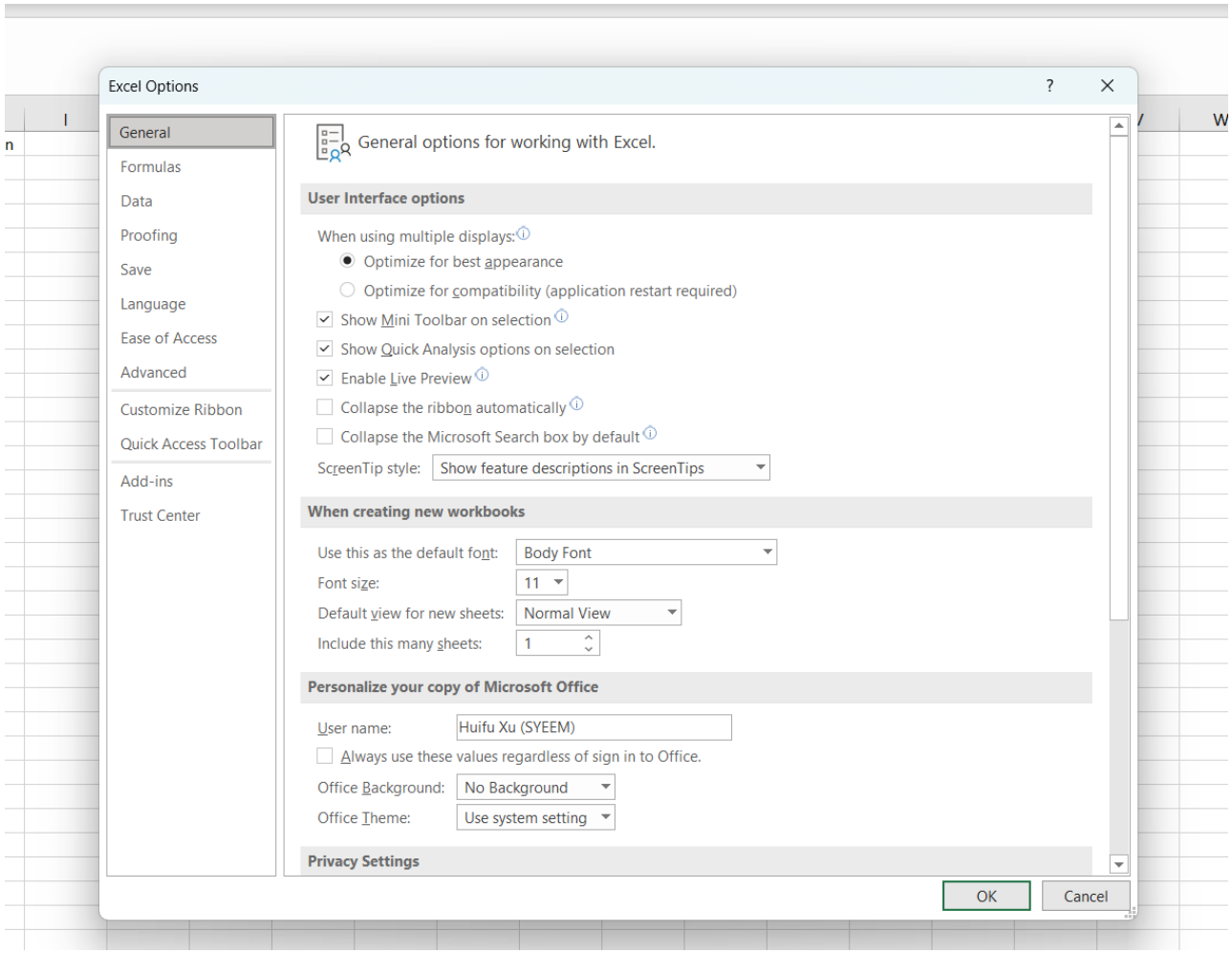
$$x_A^* = 1, \quad x_B^* = x_C^* = x_D^* = 0, PW^* = 21$$

  - This is consistent with PW analysis by choosing the project with the largest positive PW

- Chapter 6 of the course textbook.
- Chapter 13.9

Procedures to add ``LP solver'' to excel

- Go to excel, click ``file'' at top left and then ``option'' down left.
- Click add-in, choose in ``manage'' box ``excel add-ins'', click ``Go''
- Choose ``solver add-in'', click ``ok''
- Click ``data'', the ``solver'' is at top right.


Excel Options
General
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Language
Ease of Access
Advanced
Customize Ribbon
Quick Access Toolbar
Add-ins
Trust Center
General options for working with Excel.
User Interface options
When using multiple displays:
Optimize for best appearance
Optimize for compatibility (application restart required)
Show Mini Toolbar on selection
Show Quick Analysis options on selection
Enable Live Preview
Collapse the ribbon automatically
Collapse the Microsoft Search box by default
ScreenTip style:
Show feature descriptions in ScreenTips
When creating new workbooks
Use this as the default font:
Body Font
Font size:
11
Default view for new sheets:
Normal View
Include this many sheets:
1
Personalize your copy of Microsoft Office
User name:
Huifu Xu (SYEEM)
Always use these values regardless of sign in to Office.
Office Background:
No Background
Office Theme:
Use system setting
Privacy Settings
OK
Cancel

Excel Options

- General
- Formulas
- Data
- Proofing
- Save
- Language
- Ease of Access
- Advanced
- Customize Ribbon
- Quick Access Toolbar
- Add-ins
- Trust Center

View and manage Microsoft Office Add-ins.

Add-ins

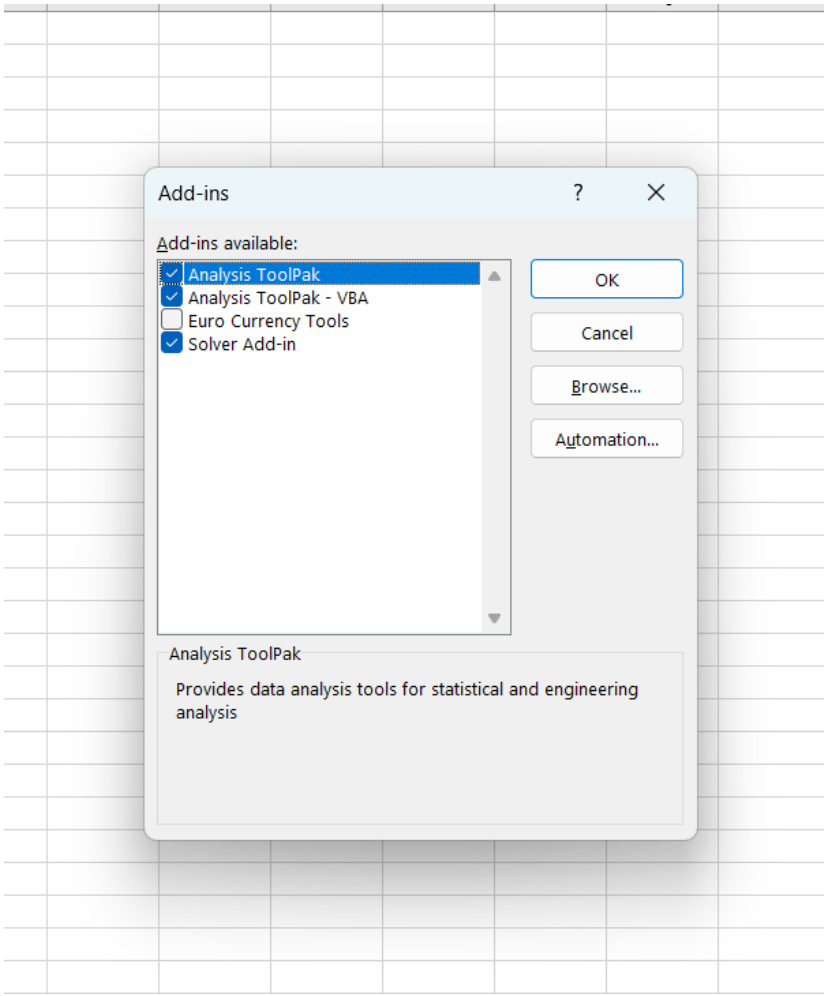
Name	Location	Type
Active Application Add-ins		
Acrobat PDFMaker Office COM Addin	C:\...x64\PDFMOfficeAddin.dll	COM Add-in
Analysis ToolPak	C:\...\Analysis\ANALYS32.XLL	Excel Add-in
Analysis ToolPak - VBA	C:\...alysis\ATPVBAEN.XLAM	Excel Add-in
Solver Add-in	C:\...\SOLVER\SOLVER.XLAM	Excel Add-in
Team Foundation Add-in	C:\...d64\TFSOfficeAdd-in.dll	COM Add-in
Visual Studio Tools for Office Design-Time Adaptor for Excel	C:\...64\VSTOExcelAdaptor.dll	COM Add-in
中文繁簡轉換增益集	C:\...ADDINS\TCSCCONV.DLL	COM Add-in
Inactive Application Add-ins		
Date (XML)	C:\...red\Smart Tag\MOFL.DLL	Action
Euro Currency Tools	C:\...ibrary\EUROTOOL.XLAM	Excel Add-in
Inquire	C:\...ice16\DCF\NativeShim.dll	COM Add-in
Microsoft Actions Pane 3		XML Expansion Pack
Microsoft Power Map for Excel	C:\...EXCELPLUGINSHELL.DLL	COM Add-in
Microsoft Power Pivot for Excel	C:\...ivotExcelClientAddIn.dll	COM Add-in

Add-in: Acrobat PDFMaker Office COM Addin
Publisher: Adobe Inc.
Compatibility: No compatibility information available
Location: C:\Program Files (x86)\Adobe\Acrobat 2017\PDFMaker\Office\x64\PDFMOfficeAddin.dll

Description: Acrobat PDFMaker Office COM Addin

Manage: Excel Add-ins Go...

OK Cancel


Add-ins
Add-ins available:
Analysis ToolPak
Analysis ToolPak - VBA
Euro Currency Tools
Solver Add-in
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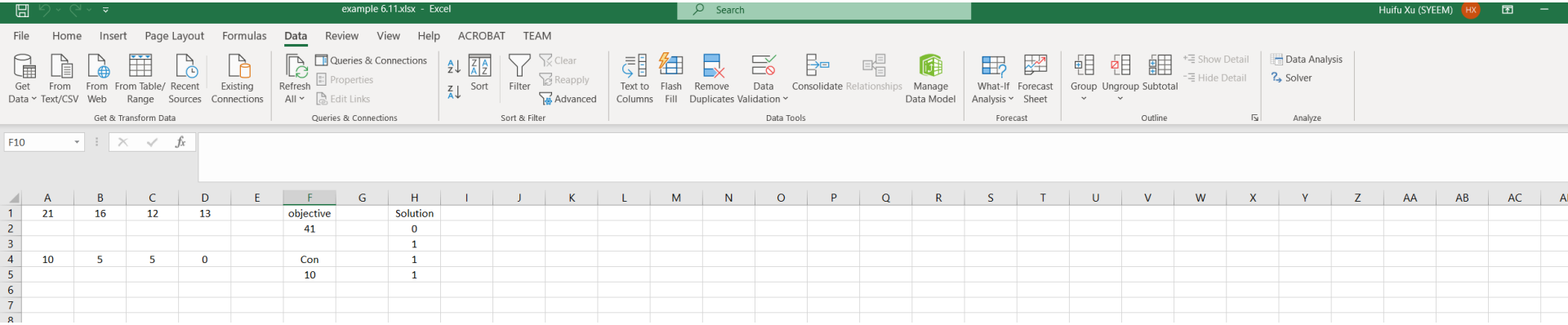
Automation...
Analysis ToolPak
Provides data analysis tools for statistical and engineering analysis


example 6.11.xlsx - Excel
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Manage Data Model
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Forecast Sheet
Group
Ungroup
Subtotal
Show Detail
Hide Detail
Data Analysis
Solver
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Forecast
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Analyze
F10
21 16 12 13
objective
Solution
41
0
1
10 5 5 0
Con
1
10
1

Procedures to solve LP

- Enter data coefficient vectors of objective and coefficients in lows
- Enter solutions in a column (all set 0)
- Use ``MMULT'' function to set objective function and constraint functions
- Click solver
- Set objective
- Set variables
- Add constraints one by one
- Set ``binary'' for all variables
- Choose ``Simplex'' as a solver

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H40

	A	B	C	D	E	F	G	H
1	21	16	12	13		objective		solution
2						0		0
3								0
4	10	5	5	0	10	con1		0
5						0		0
6	-30	-5	-5	40	10	con2		
7						0		
8	1	0	0	1	1	con3		
9						0		

Solver Parameters

Set Objective: $H$40

To: Max Min Value Of: 0

By Changing Variable Cells:

Subject to the Constraints:

Add
Change
Delete
Reset All
Load/Save

Make Unconstrained Variables Non-Negative

Select a Solving Method: GRG Nonlinear

Options

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Help Solve Close

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H2

	A	B	C	D	E	F	G	H
1	21	16	12	13		objective		solution
2						0		0
3								0
4	10	5	5	0	10	con1		0
5						0		0
6	-30	-5	-5	40	10	con2		
7						0		
8	1	0	0	1	1	con3		
9						0		

Solver Parameters

Set Objective: $H$40

To: Max Min Value Of: 0

By Changing Variable Cells:

$H$2:$H$5

Subject to the Constraints:

Add

Change

Delete

Reset All

Load/Save

Make Unconstrained Variables Non-Negative

Select a Solving Method: GRG Nonlinear

Options

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

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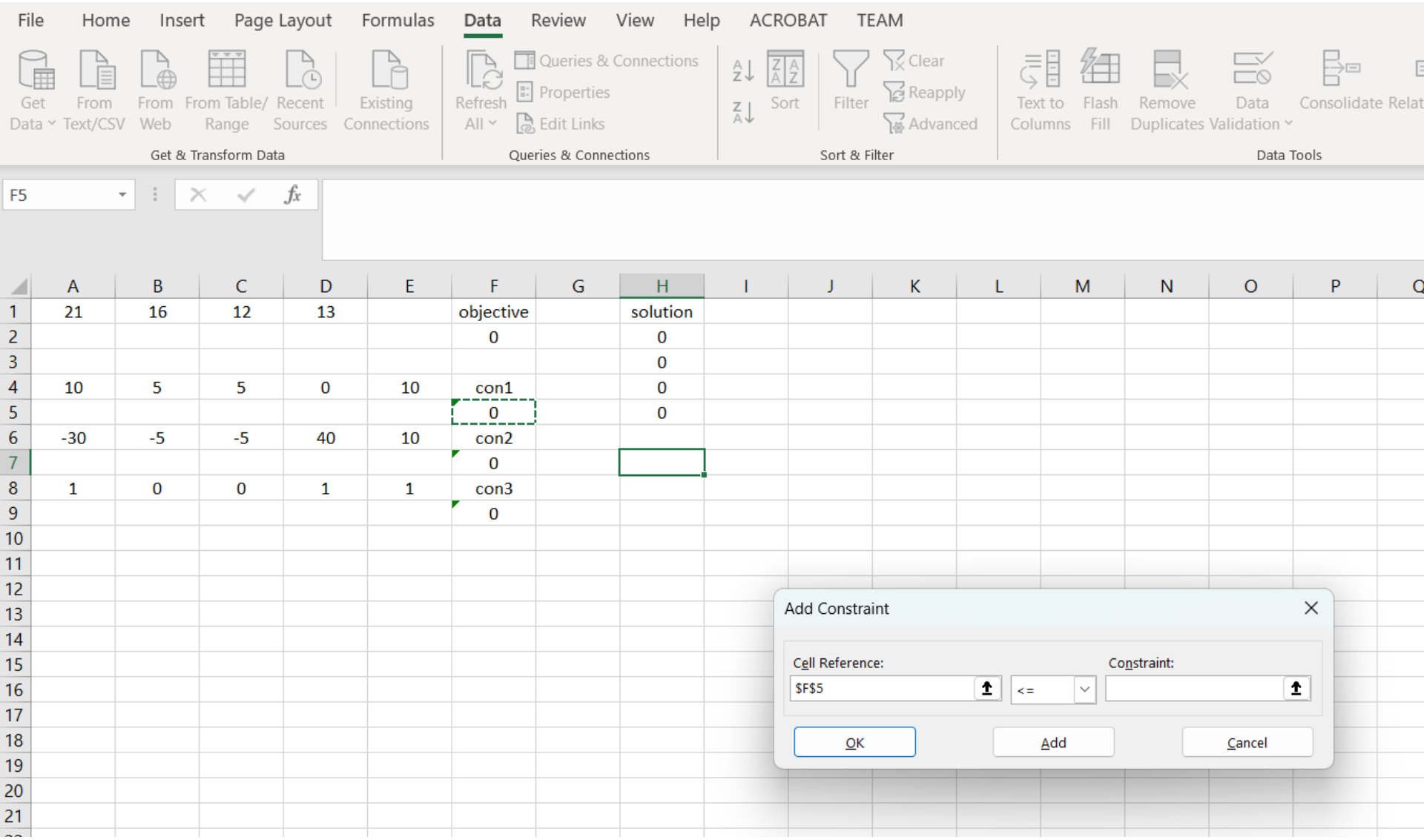
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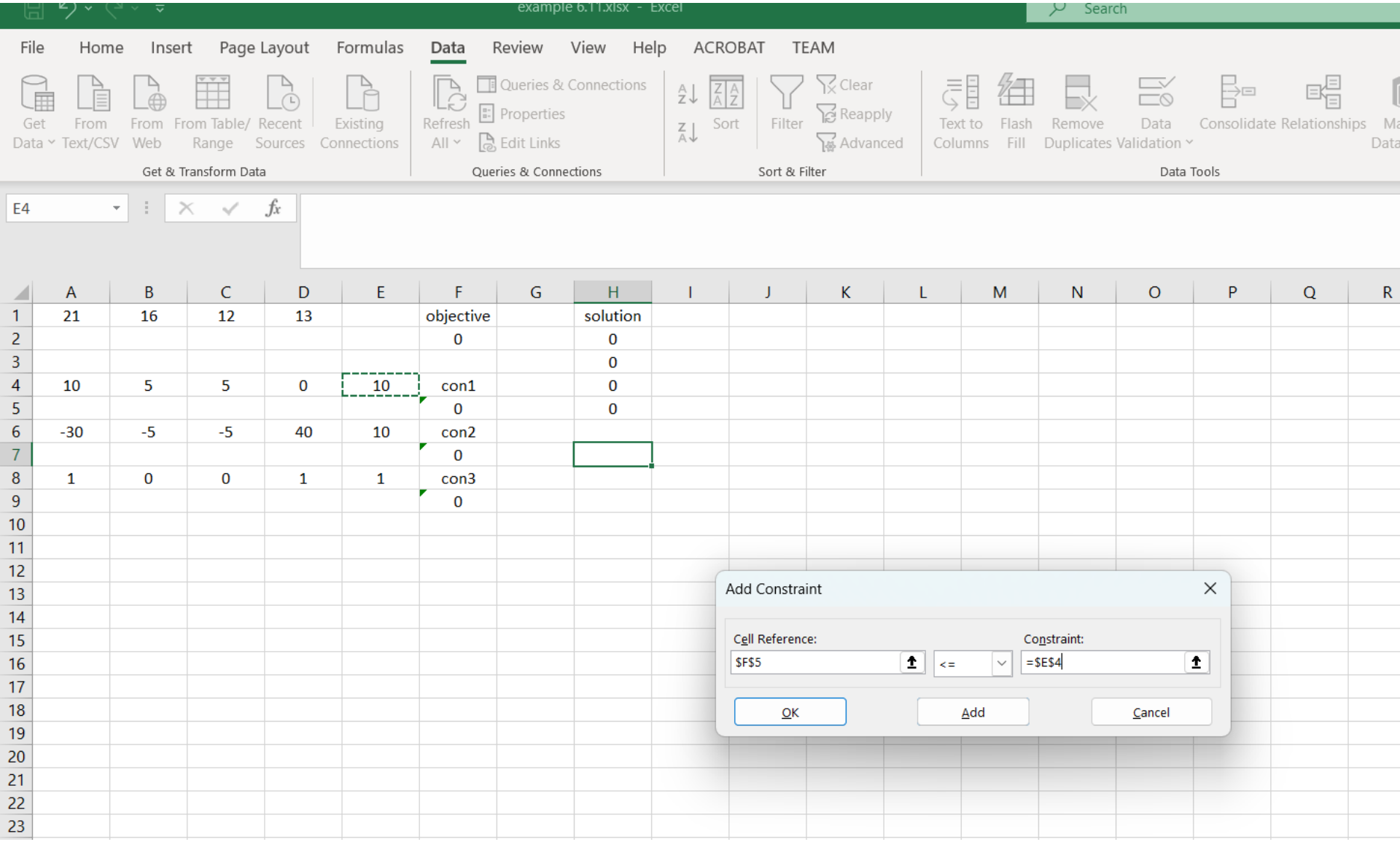
Data Tools
F5
A B C D E F G H I J K L M N O P Q
1 21 16 12 13 objective solution
2 0 0
3 0
4 10 5 5 0 10 con1 0
5 0 0
6 -30 -5 -5 40 10 con2
7 0
8 1 0 0 1 1 con3
9 0
Add Constraint
Cell Reference:
$F$5
<=
Constraint:
OK
Add
Cancel



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1	21	16	12	13		objective		solution
2						0		0
3								0
4	10	5	5	0	10	con1		0
5						0		0
6	-30	-5	-5	40	10	con2		
7						0		
8	1	0	0	1	1	con3		
9						0		

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E6

	A	B	C	D	E	F	G	H
1	21	16	12	13		objective		solution
2						0		0
3								0
4	10	5	5	0	10	con1		0
5						0		0
6	-30	-5	-5	40	10	con2		
7						0		
8	1	0	0	1	1	con3		
9						0		

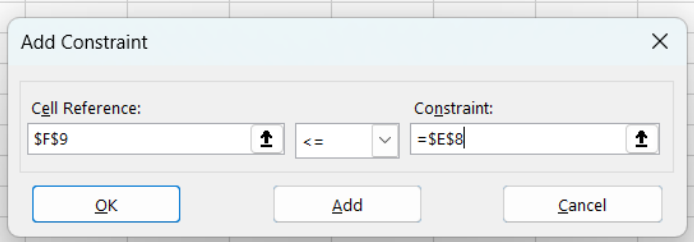
Add Constraint

Cell Reference: $F$7 | <= | Constraint: =$E$6

OK Add Cancel

E8

	A	B	C	D	E	F	G	H
1	21	16	12	13		objective		solution
2						0		0
3								0
4	10	5	5	0	10	con1		0
5						0		0
6	-30	-5	-5	40	10	con2		
7						0		
8	1	0	0	1	1	con3		
9						0		



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Get & Transform Data | Queries & Connections | Sort & Filter | Data Tools

H2

	A	B	C	D	E	F	G	H
1	21	16	12	13		objective		solution
2						0		0
3								0
4	10	5	5	0	10	con1		0
5						0		0
6	-30	-5	-5	40	10	con2		
7						0		
8	1	0	0	1	1	con3		
9						0		

Add Constraint

Cell Reference: $H$2:$H$5

<=

Constraint:

OK Add Cancel

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H7

	A	B	C	D	E	F	G	H
1	21	16	12	13		objective		solution
2						0		0
3								0
4	10	5	5	0	10	con1		0
5						0		0
6	-30	-5	-5	40	10	con2		
7						0		
8	1	0	0	1	1	con3		
9						0		

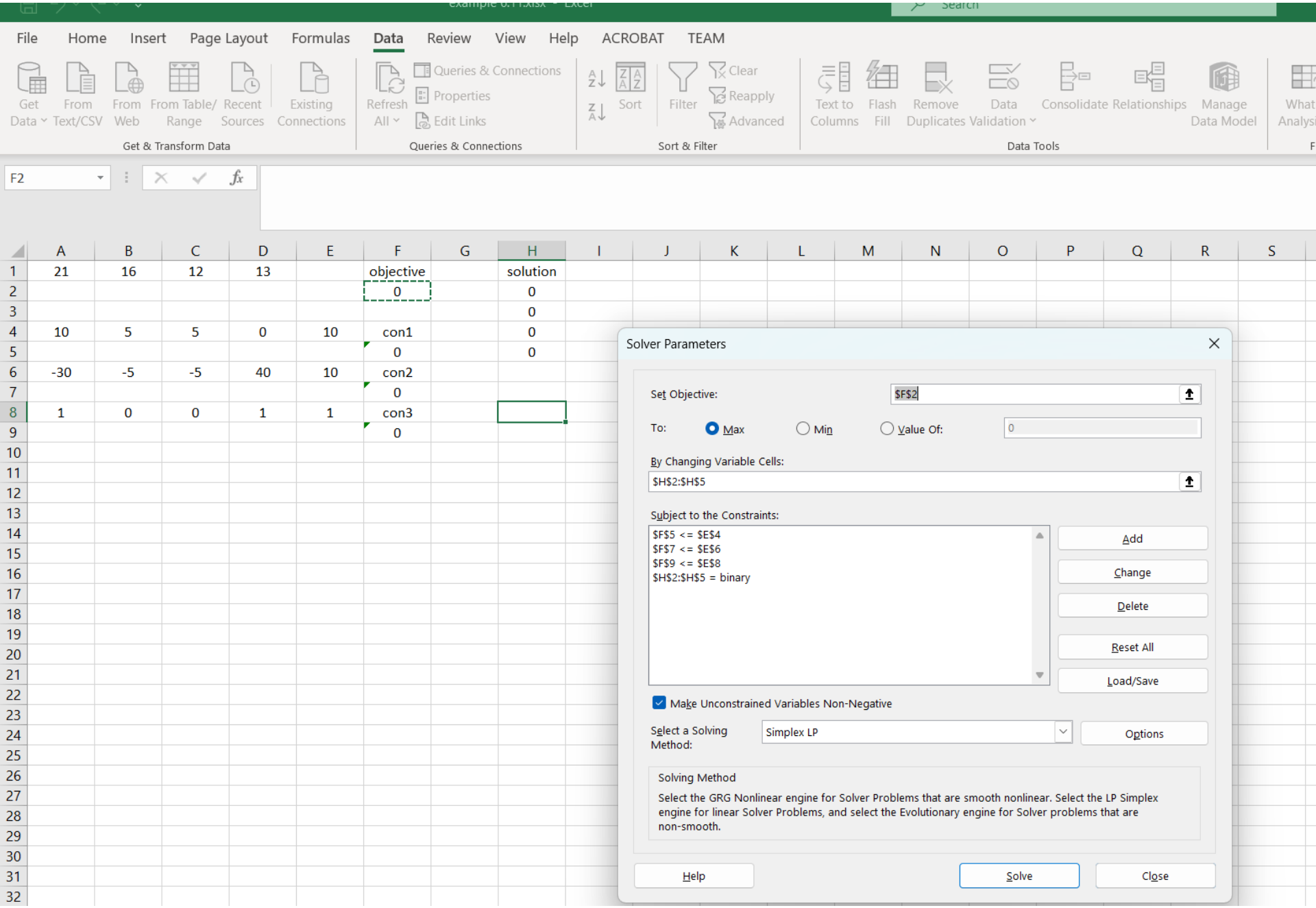
Add Constraint

Cell Reference: $H$2:$H$5

bin

Constraint: binary

OK | Add | Cancel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Data Validation
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Relationships
Manage Data Model
Get & Transform Data
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Data Tools
F2
21 16 12 13 objective solution
0 0
0
10 5 5 0 10 con1 0
0 0
-30 -5 -5 40 10 con2
0
1 0 0 1 1 con3
0
Solver Parameters
Set Objective: $F$2
To: Max Min Value Of: 0
By Changing Variable Cells:
$H$2:$H$5
Subject to the Constraints:
$F$5 <= $E$4
$F$7 <= $E$6
$F$9 <= $E$8
$H$2:$H$5 = binary
Add
Change
Delete
Reset All
Load/Save
Make Unconstrained Variables Non-Negative
Select a Solving Method:
Simplex LP
Options
Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.
Help
Solve
Close

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Get & Transform Data | Queries & Connections | Sort & Filter | Data Tools | Forecast

H8

	A	B	C	D	E	F	G	H
1	21	16	12	13		objective		solution
2						28		0
3								1
4	10	5	5	0	10	con1		1
5						10		0
6	-30	-5	-5	40	10	con2		
7						-10		
8	1	0	0	1	1	con3		
9						0		

Solver Results

Solver found a solution. All Constraints and optimality conditions are satisfied.

Reports

Answer

Keep Solver Solution

Restore Original Values

Return to Solver Parameters Dialog

Outline Reports

OK | Cancel | Save Scenario...

Solver found a solution. All Constraints and optimality conditions are satisfied.

When the GRG engine is used, Solver has found at least a local optimal solution. When Simplex LP is used, this means Solver has found a global optimal solution.